MAT 221 - CALCULUS I
Lesson 10-11: - Improper Integrals

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Improper Integrals of Type I

Integrals with infinite limits of integration are **improper integrals of type I**

1. If $f(x)$ is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx$$

2. If $f(x)$ is continuous on $(-\infty, b]$, then

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty} \int_a^b f(x)\,dx$$

3. If $f(x)$ is continuous on $(-\infty, \infty)$, then

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^{\infty} f(x)\,dx$$

where c is any real number.

- In each case, if the limit is finite we say that the improper integral **converges** and that the limit is **value** of the improper integral.
- If the limit fails to exist, the improper integral **diverges**.

Example 1

Determine if the integral is convergent or divergent.

(a) $\int_0^{\infty} e^{-x}\,dx$ (b) $\int_{-\infty}^{-1} \frac{4}{x^2}\,dx$ (c) $\int_{-\infty}^{\infty} \frac{4x^3}{(x^4+1)^2}\,dx$ (d) $\int_{-1}^{\infty} x\,dx$

Solution

(a) $\int_0^{\infty} e^{-x}\,dx = \lim_{b\to\infty} \int_0^{b} e^{-x}\,dx = \lim_{b\to\infty} (-e^{-x})|_0^b = \lim_{b\to\infty} [-e^{-b}+1] = 1$

$\therefore$ the integral converges to 1.

(b) $\int_{-\infty}^{-1} \frac{4}{x^2}\,dx = \lim_{a\to-\infty} \int_a^{-1} \frac{4}{x^2}\,dx = \lim_{a\to-\infty} \left[\frac{-4}{x}\right]_a^{-1} = 4$

$\therefore$ the integral converges to 4.

(c)

$$\int_{-\infty}^{\infty} \frac{4x^3}{(x^4+1)^2}\,dx = \lim_{a\to-\infty} \int_a^{0} \frac{4x^3}{(x^4+1)^2}\,dx + \lim_{b\to\infty} \int_0^{b} \frac{4x^3}{(x^2+1)^2}\,dx$$

$$= \lim_{a\to-\infty} \left[\frac{-1}{x^4+1}\right]_a^{0} + \lim_{b\to\infty} \left[\frac{-1}{x^4+1}\right]_0^{b} = 0$$

(d) $\int_{-1}^{\infty} x\,dx = \lim_{b\to\infty} \int_{-1}^{b} x\,dx = \lim_{b\to\infty} \left[\frac{x^2}{2}\right]_{-1}^{b} = \lim_{b\to\infty} \left[\frac{b^2}{2} - \frac{1}{2}\right] = \infty$

$\therefore$ the integral diverges.

Integrals of the form $\int_1^\infty \frac{1}{x^p}dx$

The function $f(x) = \frac{1}{x}$ is the boundary between the convergent and divergent improper integrals with integrands of the form $y = \frac{1}{x^p}$. The improper integral converges if $p > 1$ and diverges if $p \leqslant 1$.

Example 2

Find the values of p for which $\int_1^\infty \frac{1}{x^p}dx$ converges.

Solution

If $p \neq 1$,

$$\int_1^\infty \frac{1}{x^p}dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^p}dx = \lim_{b\to\infty}\left[\frac{1}{1-p}\left(\frac{1}{b^{p-1}} - 1\right)\right] = \begin{cases} \frac{1}{p-1}, & \text{if } p > 1; \\ \infty, & \text{if } p > 1. \end{cases}$$

$\therefore \quad \int_1^\infty \frac{1}{x^p}dx$ converges if $p > 1$ and diverges if $p < 1$.

If $p = 1$, the integral also diverges

$$\int_1^\infty \frac{1}{x^p}dx = \int_1^\infty \frac{1}{x}dx = \lim_{b\to\infty}\int_1^b \frac{1}{x}dx = \lim_{b\to\infty} \ln x|_1^b = \lim_{b\to\infty}(\ln b - \ln 1) = \infty$$

Improper Integrals of Type II

- Another type of improper integral arises when the integrand has a vertical asymptote - an infinite discontinuity- at a limit of integration or at some point between the limits of integration.
- Integrals of functions that become infinite at a point within the interval of integration are **improper integrals of Type II**
  1. If $f(x)$ is continuous on $(a, b]$ and discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{c \to a^+} \int_c^b f(x)\,dx.$$

  2. If $f(x)$ is continuous on $[a, b)$ and discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{c \to b^-} \int_a^c f(x)\,dx.$$

  3. If $f(x)$ is discontinuous at c, where $a < c < b$, and continuous on $[a, c) \cup (c, b]$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Example 3

Determine if the integral is convergent or divergent.

(a) $\int_0^1 \frac{dx}{x}$ (b) $\int_0^1 \frac{dx}{1-x}$ (c) $\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$ (d) $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$

Solution

(a) We note $f(x) = \frac{1}{x}$ is discontinuous at $x = 0$. Then

$$\int_0^1 \frac{dx}{x} = \lim_{a\to 0^+} \int_a^1 \frac{dx}{x} = \lim_{a\to 0^+} \ln x|_a^1 = \lim_{a\to 0^+} [\ln 1 - \ln a]$$
$$= \lim_{a\to 0^+} [-\ln a] = \infty$$

Integral diverges

(b) $f(x) = \frac{1}{1-x}$ is discontinuous at $x = 1$. Then

$$\int_0^1 \frac{dx}{1-x} = \lim_{b\to 1^-} \int_0^b \frac{dx}{1-x} = \lim_{b\to 1^-} [-\ln(1-x)]|_0^b$$
$$= \lim_{a\to 1^-} [-\ln(1-b) - \ln 1] = \lim_{b\to 1^-} [-\ln(1-b)]$$
$$= \infty$$

Integral diverges

(c) We note $f(x) = \frac{1}{(x-1)^{\frac{2}{3}}}$ is discontinuous at $x = 1$. So that

$$\begin{aligned}
\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}} &= \int_0^1 \frac{dx}{(x-1)^{\frac{2}{3}}} + \int_1^3 \frac{dx}{(x-1)^{\frac{2}{3}}} \\
&= \lim_{b\to 1^-} \int_0^b \frac{dx}{(x-1)^{\frac{2}{3}}} + \lim_{a\to 1^+} \int_a^3 \frac{dx}{(x-1)^{\frac{2}{3}}} \\
&= \lim_{b\to 1^-} \left[3(b-1)^{\frac{1}{3}}\right]_0^b + \lim_{a\to 1^+} \left[3(b-1)^{\frac{1}{3}}\right]_0^b \\
&= 3 + 3(2)^{\frac{1}{3}}
\end{aligned}$$

Integral converges

(d) We choose $c = 0$ as a break point of the integral so that

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{dx}{1+x^2} &= \int_{-\infty}^{0} \frac{dx}{1+x^2} + \int_0^{\infty} \frac{dx}{1+x^2} \\
&= \lim_{a\to -\infty} \int_a^0 \frac{dx}{1+x^2} + \lim_{b\to \infty} \int_0^{\infty} \frac{dx}{1+x^2} \\
&= \lim_{a\to -\infty} [\tan^{-1} x]|_a^0 + \lim_{b\to \infty} [\tan^{-1} x]|_0^b \\
&= \frac{\pi}{2} + \frac{\pi}{2} = \pi.
\end{aligned}$$